

1. Work through the derivation of the two point function at order λ in a $\lambda\phi^4$ theory, that was discussed in class. Complete the calculation which was introduced in the lecture., deriving the final result using the functional method.
2. Consider the loop diagrams of a scalar field theory defined by the Lagrangian:

$$L = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4, -\frac{g}{3!}\phi^3.$$

Write down the expressions for all the 1PI loop diagrams (with external propagators amputated) shown in Figure 1, with the correct symmetry factors. Please write down the Feynman rules you use. You do not need to evaluate any of the loop integrals.

3. Next consider both scalar and spinor QED. The Lagrangians were discussed in 513 and are given in Peskin and Schroder. Write down the expression for all the 1PI loop diagrams (external propagators amputated) shown in Figure 2, with correct symmetry factors. Note that the first diagram is in scalar QED. Others are all in spinor QED. Again you do not need to evaluate the loop integrals.